Tribhuvan University

Faculty of Humanities and Social Sciences

Paathshala – A web based learning management system

A PROJECT REPORT

Submitted to

Department of Computer Application

Nepathya College

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by Anish Paudel and Niruta Gharti entitled "**Paathshala"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

Signature of Supervisor

Sushant Malla

……

Tribhuvan University

Faculty of Humanities and Social Science

Nepathya College

LETTER OF APPROVAL

This is to certify that this project prepared by Anish Paudel and Niruta Gharti entitled "**Paathshala"** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

Signature of Supervisor	**Signature of HOD/ Coordinator**
Signature of Internal Examiner	**Signature of External Examiner**

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List of Abbreviations

CRUD: Create, Read, Update, Delete

ER-Model: Entity-Relationship Model

DFD: Data Flow Diagram

UI: User Interface

API: Application Programming Interface

JWT: JSON Web Token

IDE: Integrated Development Environment

HTML: HyperText Markup Language

CSS: Cascading Style Sheets

SQL: Structured Query Language

JSON: JavaScript Object Notation

LLM: Large Language Model

AI: Artificial Intelligence

Chapter 1: Introduction

1.1 Introduction

The advancement of digital technology has transformed education by enabling flexible and accessible online learning. However, in countries like Nepal, many existing e-learning platforms lack affordability, simplicity, and adaptation to local needs.

This project presents Paathshala, a Learning Management System (LMS) designed to provide a structured and user-friendly online learning environment for students and professionals. The system supports course management, student enrollment, progress tracking, and MCQ-based assessments with adaptive difficulty. It also includes an AI-powered study assistant for enhanced learning support.

1.2 Problem Statement

In Nepal, the existing online education platforms lack several critical features that modern learners need. Most platforms do not provide adaptive assessment systems, AI-powered study assistance, or comprehensive analytics for both students and administrators.

The traditional education system in many regions still relies heavily on physical classrooms, limited teaching resources, and fixed schedules, which restrict accessibility and flexibility for students. In countries like Nepal, many students face challenges such as lack of quality educational materials, geographical barriers, and limited interaction with instructors outside the classroom.

1.3 Objectives

- To provide centralized learning resources that offer a single platform with well-structured notes, tutorials, quizzes, model questions for competitive exam preparation and learning.
- To enhance accessibility that ensures learners from diverse geographical regions can access quality study materials anytime, anywhere, at an affordable cost.
- To enable flexible learning that supports self-paced learning where students can revisit concepts, pause, and resume based on their schedule.
- To ensure scalability & reliability for building a robust and scalable platform capable of supporting large numbers of learners and diverse exam categories.

1.4 Scope and Limitations

Scope

- **Target Audience:** Students preparing for various competitive examinations (academic, professional, and government-based).
- **Learning Resources:** Comprehensive study materials including notes, video tutorials, recorded lectures, quizzes.
- **System Design:** Web-based platform (scalable for mobile app in the future) with secure authentication and role-based access.
- **Assessment:** Adaptive tests, analytics dashboards, and AI assistance.

Features for Learners:

- User registration, profile management, and progress tracking.
- Course browsing, enrollment, and structured learning paths.
- Practice quizzes with instant results.
- Certificates of completion after finishing courses/tests.

Features for Admins:

- Course and content creation/management.
- Performance monitoring of enrolled students.

Limitations

- **Internet Dependency:** Requires a stable internet connection; learners in low-connectivity areas may face difficulties accessing video lectures and live sessions.
- **Limited Offline Access:** Study materials and recorded sessions may not be available offline in early versions.
- **Scalability Challenges:** With a growing user base, ensuring smooth performance and server reliability may require additional infrastructure investment.
- **Human Interaction Gap:** Although tutorials help, they may not fully replicate the real-time personal guidance available in traditional classrooms.

Chapter 2: Background Study and Literature Review

2.1 Background Study

The increasing competitiveness in academic and professional examinations has created a growing demand for effective, accessible, and affordable exam preparation platforms. Traditional coaching centers, while helpful, suffer from limitations such as high costs, rigid schedules, and restricted geographic reach. Learners often resort to fragmented resources books, handwritten notes, free online videos—which lack structure, consistency, and personalization.

Paathshala aims to fill this gap by providing an integrated platform designed exclusively for competitive exam aspirants. It combines multiple learning resources—comprehensive notes, tutorials, practice tests, quizzes and live AI assessment—into a single, unified hub. By leveraging modern web technologies and cloud infrastructure, the system ensures scalability, reliability, and accessibility, offering students 24/7 learning support.

2.2 Literature Review

Learning Management Systems (LMS) play a vital role in modern education by enabling efficient course delivery, communication, and performance tracking. H. Coates, R. James, and G. Baldwin highlight that LMS platforms improve the quality of teaching and learning by providing structured content and better interaction between students and instructors [1]. Their study emphasizes that LMS enhances flexibility and accessibility in higher education.

E-learning platforms have gained significant popularity due to their ability to provide scalable and self-paced learning. According to Al-Fraihat et al., the success of e-learning systems depends on factors such as system quality, content quality, and user satisfaction [2]. These systems support diverse learners but often face challenges related to engagement and usability.

With the advancement of technology, Artificial Intelligence (AI) has been integrated into e-learning systems to improve personalization. Holmes et al. explain that AI-driven educational systems can analyze learner behavior and provide adaptive learning experiences, which improve student performance and engagement [3]. Adaptive assessment techniques dynamically adjust question difficulty, making evaluation more effective.

2.3 Methodology

The development of paathshala follows traditional waterfall methodology. Involves sequential flow of planning and requirement gathering, requirements analysis, system designing, implementation, testing, deployment of the system, maintenance support.

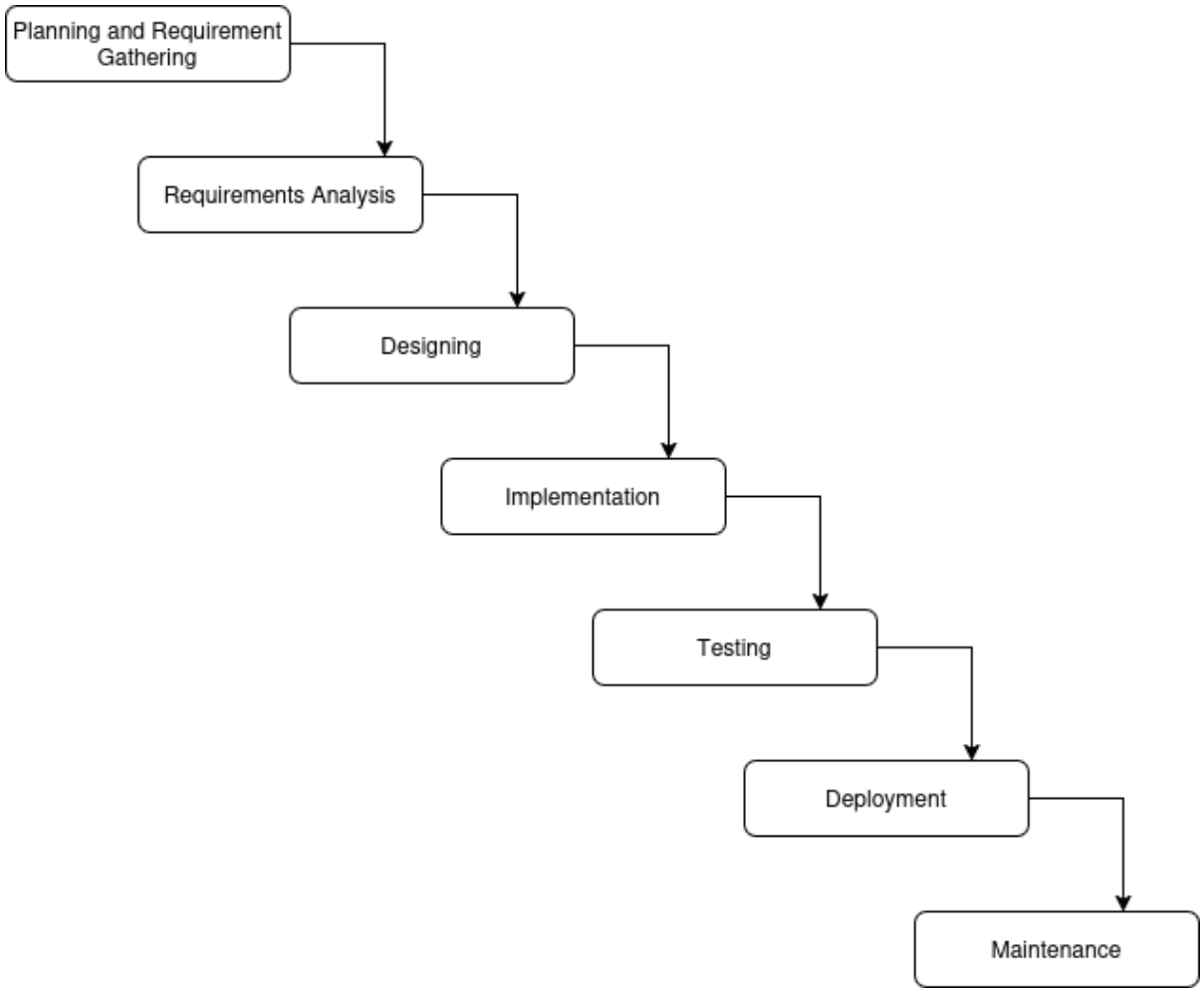

Figure 1: WaterFall Methodology

Chapter 3: System Analysis and Design

3.1 System Analysis

System Analysis for Paathshala involves examining the platform's requirements and processes to design an effective and user-friendly exam-preparation system. This process focuses on understanding the needs of learners and instructors, identifying potential challenges, and proposing improvements to optimize the learning experience. The objectives of system analysis for Paathshala include:

- Evaluating the feasibility of the platform concept in terms of usability, scalability, and technical requirements.
- Performing economic and technical analysis to ensure cost-effective and efficient development.
- Allocating functions to software modules, hardware resources, user roles, databases, and other system components.
- Establishing cost estimates, development timelines, and schedule constraints to guide implementation.

3.1.1 Requirement Analysis

Requirement analysis involves refining and prioritizing the collected requirements to ensure they align with the objectives of Paathshala and the project constraints. This process helps define what the system must do to effectively support learners and instructors. The requirements are categorized as follows:

i. Functional Requirements

- **User Authentication:** Users should be able to register, log in, and log out securely.
- **User Roles & Permissions:** The system should differentiate access for basic user , subscribed users and administrators, ensuring that each role has appropriate privileges. Basic users have access to free limited resources & subscribed users have access to wide paid resources.
- **Content Management (CRUD Operations):** Admins should be able to create, read, update, and delete notes, quizzes, tests, and tutorials.
- **Enrollment Management and Subscription:** Users can enroll in courses, modules, or learning tracks by purchasing subscriptions for individual premium courses.

- **Search Functionality:** Users can search for courses based on keywords, tags, or subjects.
- **Progress Tracking:** The platform should track users' progress in quizzes and display performance metrics.
- **Assessment & Feedback:** Users can attempt quizzes, and the system should provide automated scoring. Also users can access AI assistance.
- **Content Display:** Learning materials should be presented clearly, including formatted text, images, videos, and other media.

ii. Non Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system, ensuring that Paathshala is reliable, efficient, and user-friendly. These requirements include:

- **Performance:** The system should respond to user actions (e.g., page loads, searches, quiz submissions) within 2–3 seconds under normal load conditions.
- **Scalability:** The platform should handle increasing numbers of users and content without performance degradation.
- **Security:** User data, including personal details and assessment results, must be securely stored and transmitted using encryption standards enforcing role-based access control.
- **Usability:** The system should have an intuitive interface, allowing users of varying technical skill levels to navigate easily. Accessibility standards should be followed to support users with disabilities.
- **Reliability & Availability:** The platform should ensure minimal downtime and be available 24/7, with backup and recovery mechanisms in place.
- **Maintainability:** The system should be modular and well-documented, allowing developers to easily update, fix, or enhance features.
- **Compatibility:** The platform should be compatible with major web browsers (Chrome, Firefox, Edge, Safari) and adapt to different screen sizes (desktop, tablet, mobile).
- **Data Integrity:** All user actions, progress, and assessment data should be accurately recorded and preserved without loss or corruption.

3.1.2 Feasibility Analysis

Feasibility study assesses whether the development and implementation of Paathshala are practical, achievable, and worthwhile. It evaluates the project from multiple perspectives to ensure that the platform can be successfully delivered within constraints. The study includes the following aspects:

i. Technical Feasibility

This assesses whether the technology and resources required to develop Paathshala are available and sufficient. The platform relies on web technologies such as React for the frontend, Spring Boot for the backend, and cloud storage/services for content management. The team has the necessary expertise to implement these technologies, ensuring that the system can be developed, maintained, and scaled effectively.

ii. Operational Feasibility

Operational feasibility examines whether the platform will function as intended in the real-world environment. Paathshala is designed to meet the needs of students and instructors, with features like content management, quizzes, live sessions, and progress tracking. User-centered design and iterative testing ensure that the system will be easy to use, practical, and aligned with educational workflows.

iii. Economic Feasibility

Economic feasibility evaluates the cost-effectiveness of the project. The development costs, including software, hardware, cloud services, and personnel, are compared against the expected benefits, such as improved learning outcomes, accessibility, and potential revenue streams. Preliminary analysis indicates that the project is financially viable and justifiable.

iv. Schedule:

This assesses whether the project can be completed within a reasonable timeframe. The development process is planned in phases, including requirement gathering, design, implementation, testing, and deployment. With a structured timeline and iterative approach, the project is feasible to complete within the allocated schedule. The project is estimated to be completed in between 14-18 weeks. The Gantt chart is as follow:

Project Gantt Chart - 12 Weeks

#	Phase	Start	End	1	2	3	4	5	6	7	8	9	10	11	12
1	Planning	1	2												
2	Analysis	2	3												
3	Designing	4	5												
4	Implementation	6	9												
5	Testing	9	11												
6	Deployment	11	12												
7	Documentation	11	12												

** Shaded cells indicate active weeks for each phase*

Figure 2: Gantt Chart

3.1.3 Data Modelling (ER-Modeling)

Data modeling is an essential step in designing the database structure for Paathshala, ensuring proper organization, storage, and retrieval of learning management system (LMS) data. The Entity-Relationship (ER) model provides a clear visualization of the system's main entities, their attributes, and relationships.

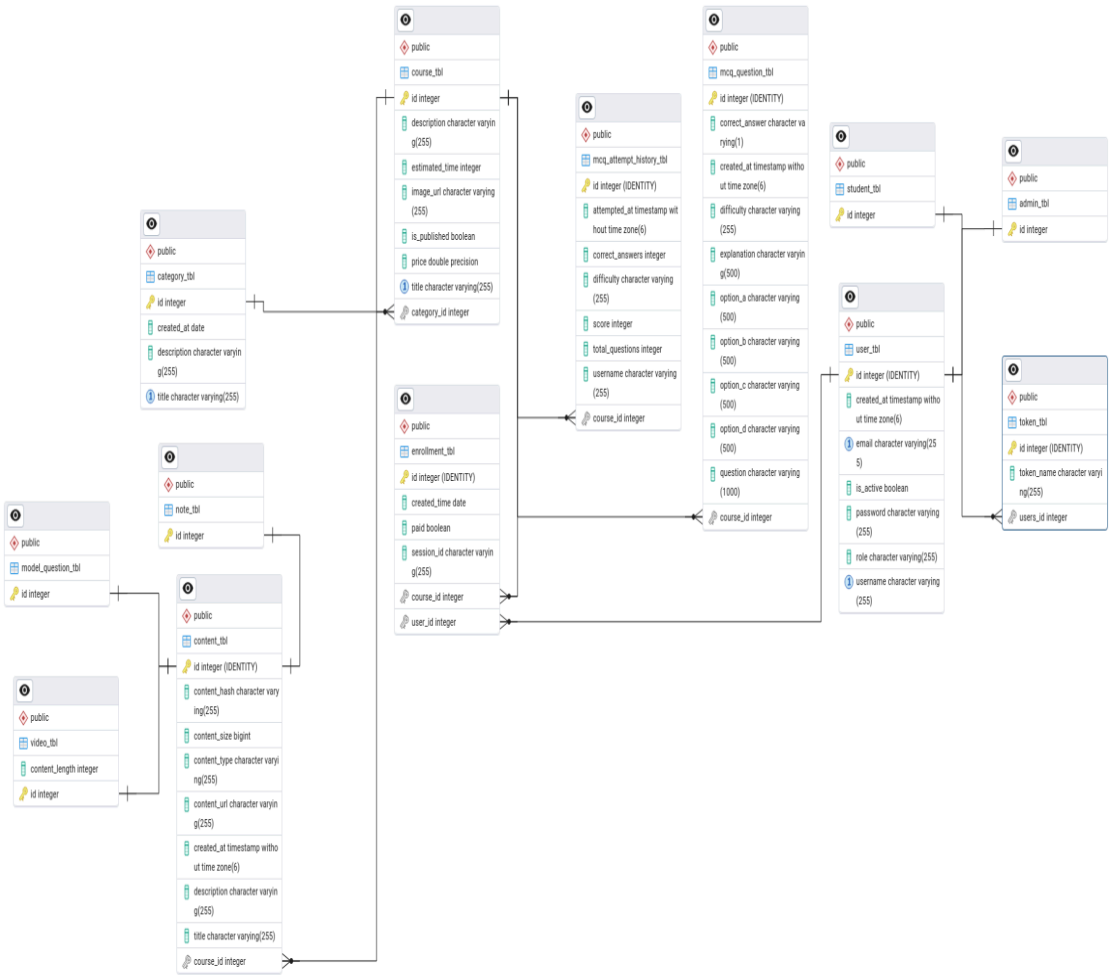

Figure 3: Er Model

3.1.4 Process Modeling (DFD)

Process Modeling with DFDs visually represents data flow in a system, showing processes, data stores, and external entities. Level 1 breaks it into major processes, and Level 2 details each process. They help understand, design, and document systems effectively.

Level 1 DFD breaks the system into major processes: User Management, Admin Management, Quiz Management,etc showing how data flows between the user, processes, and the database.

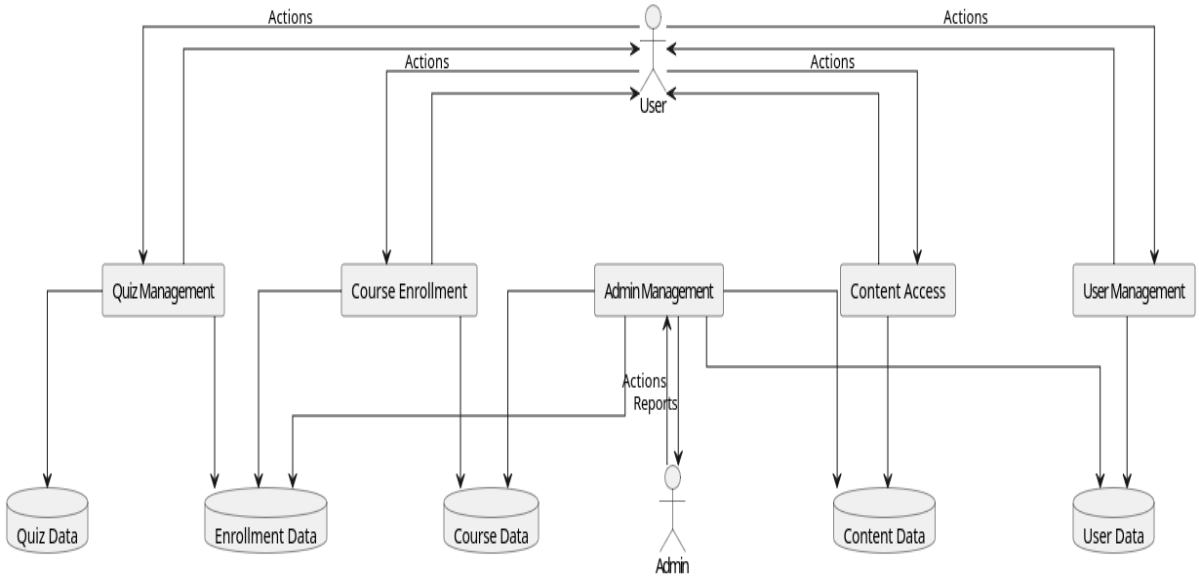

Figure 4: Level 1 DFD

Level 2 DFD decomposes each process into detailed steps, such as browsing courses, selecting courses, and subscription payment, while maintaining data flow between the user and the database.

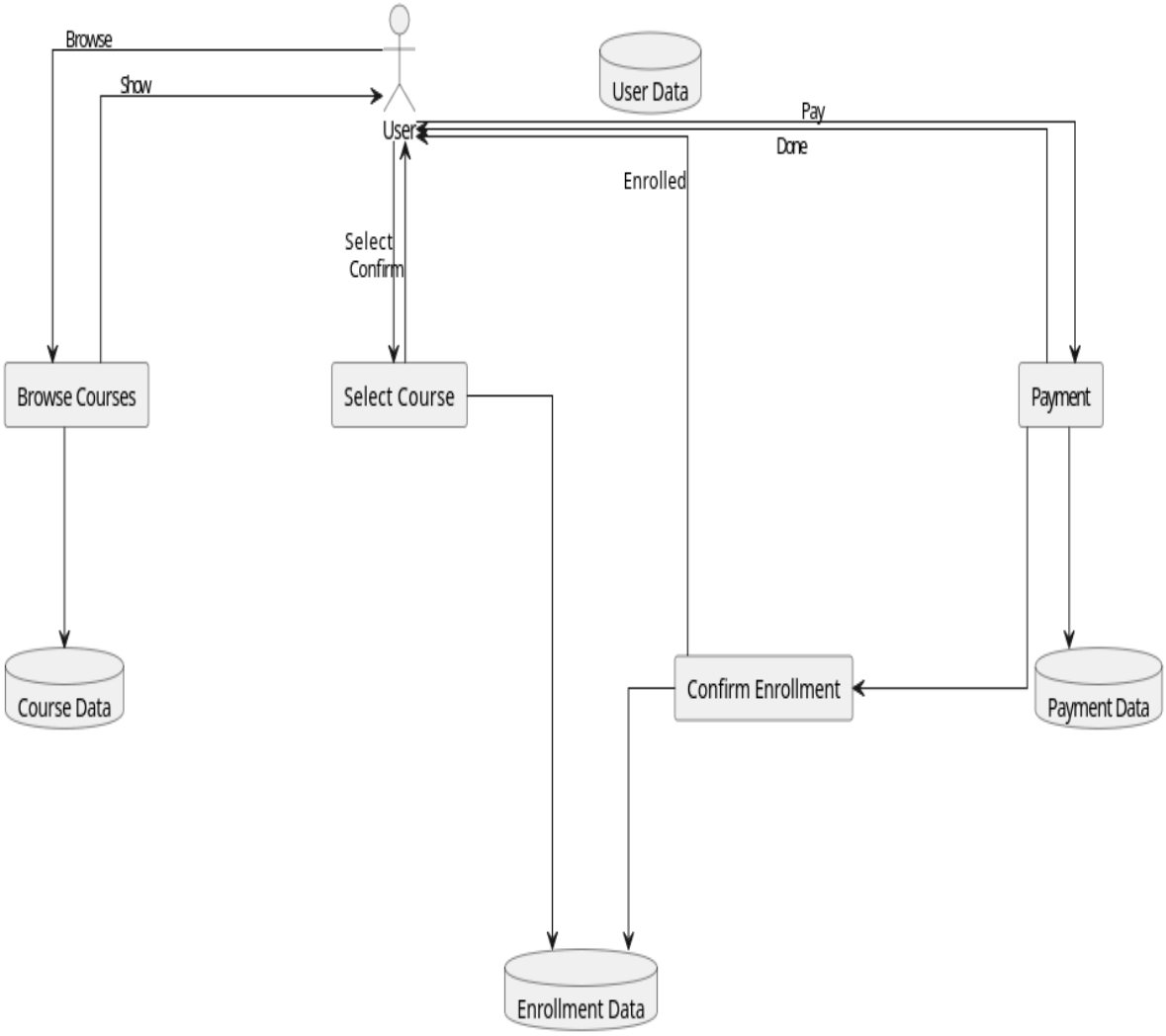

Figure 5: Level 2 DFD

3.2 System Design

3.2.1. Architectural Design

Architectural design is a process for identifying the sub-systems making up a system and the framework for sub-system control and communication.

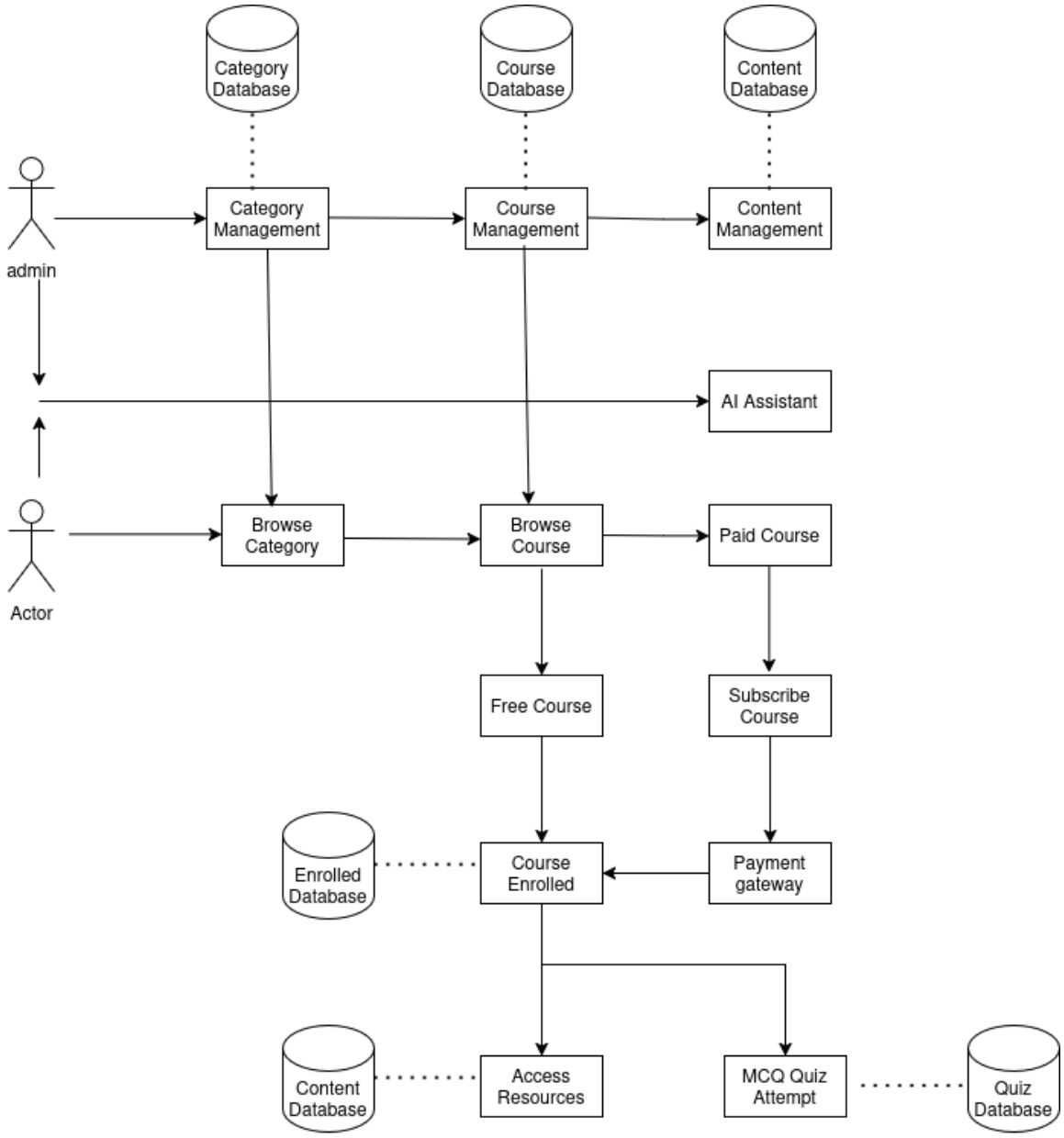

Figure 6: Architectural Design

3.2.2 Flow Chart

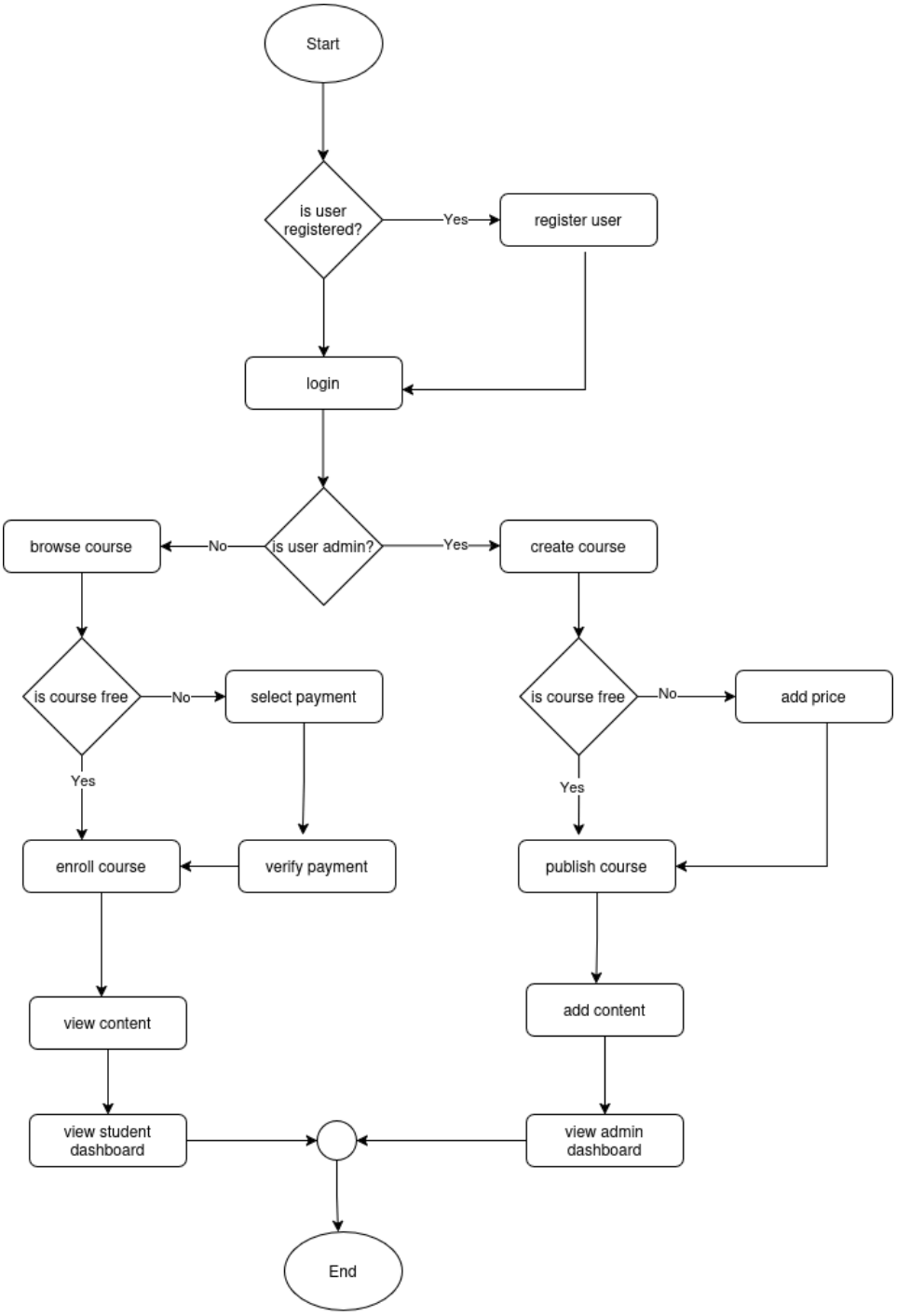

Figure 7: Flowchart

3.2.3 Sequence Diagram

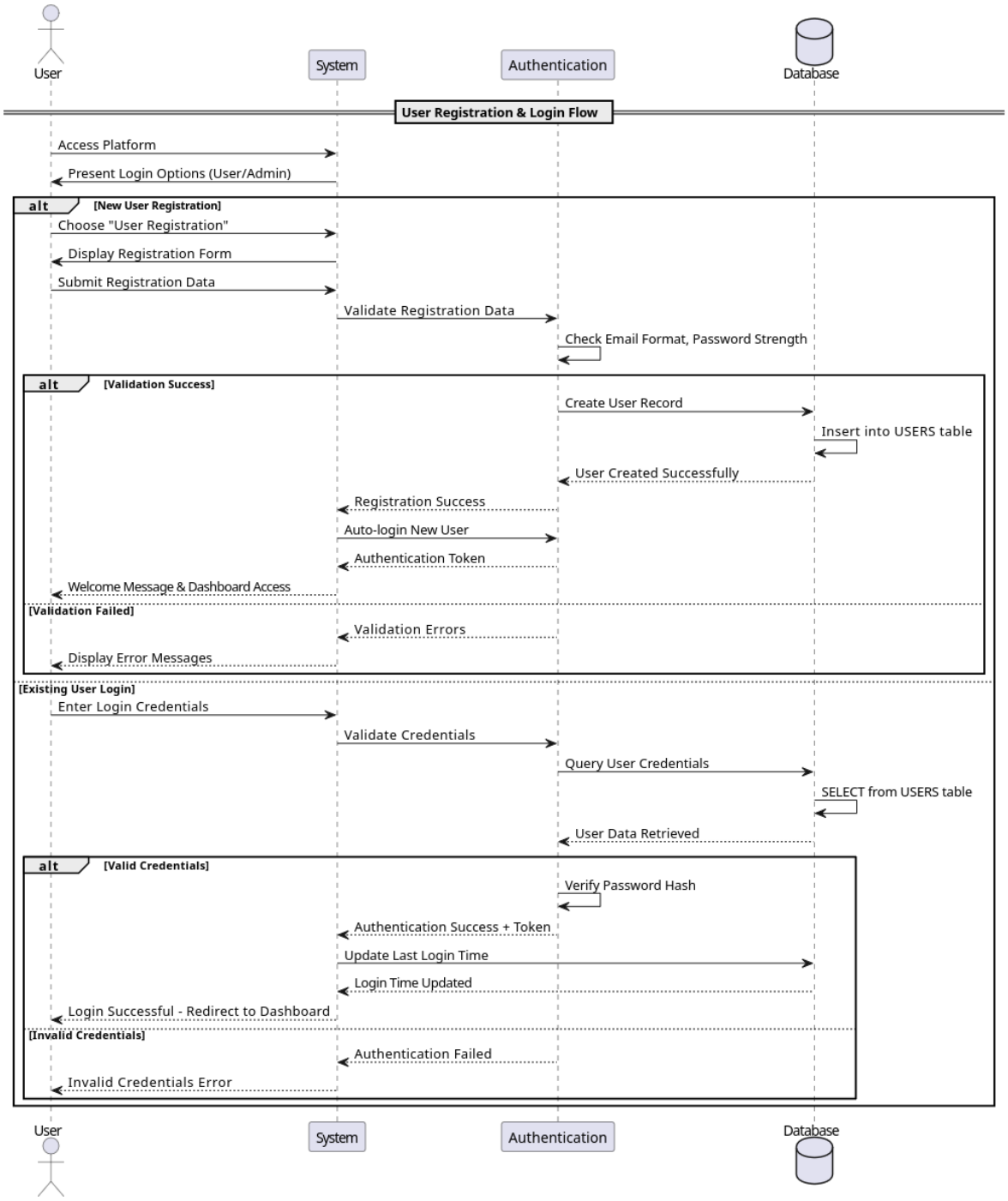

Figure 8: User Registration and Authentication Flow

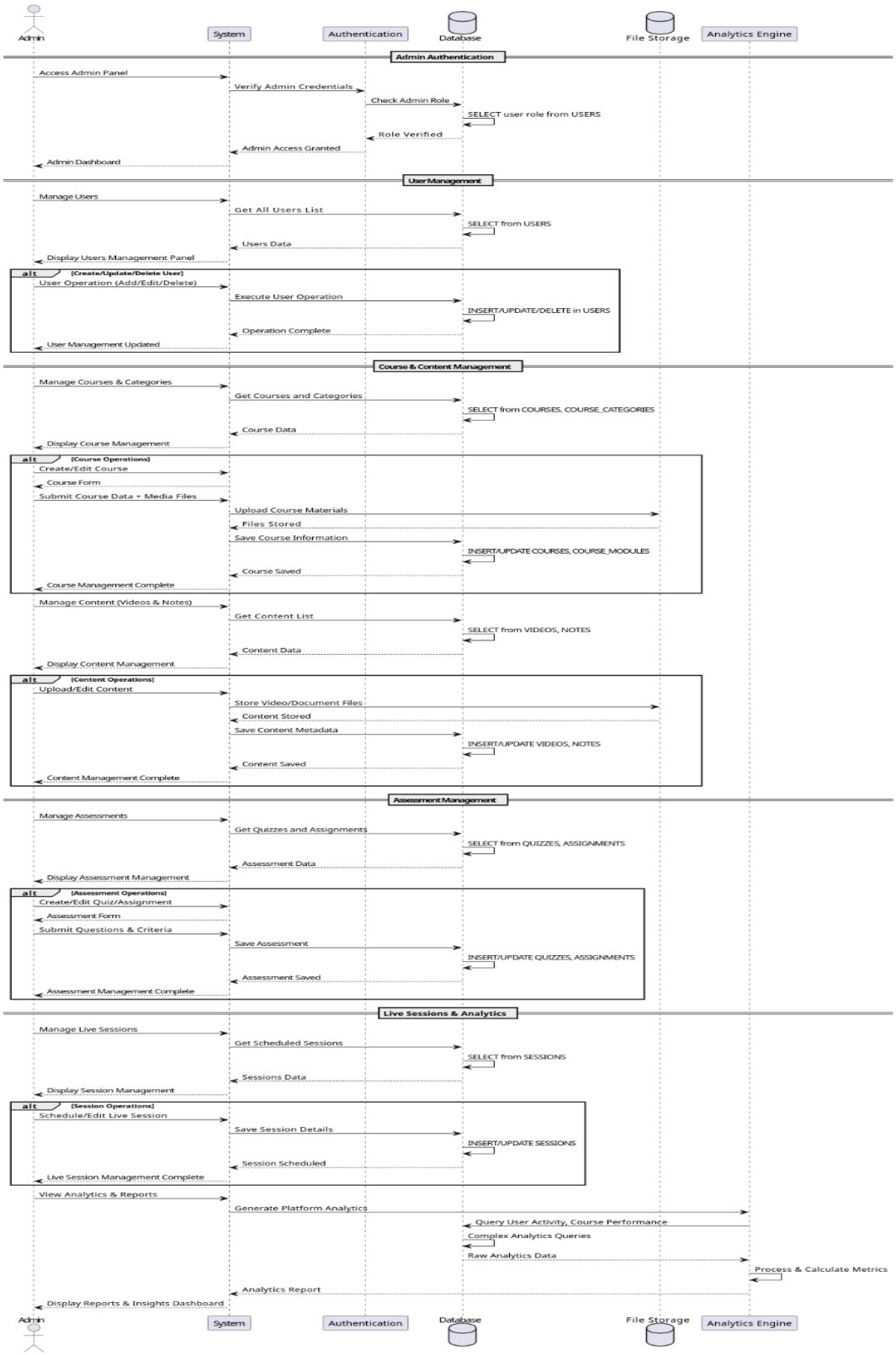

Figure 9: Admin Management Flow

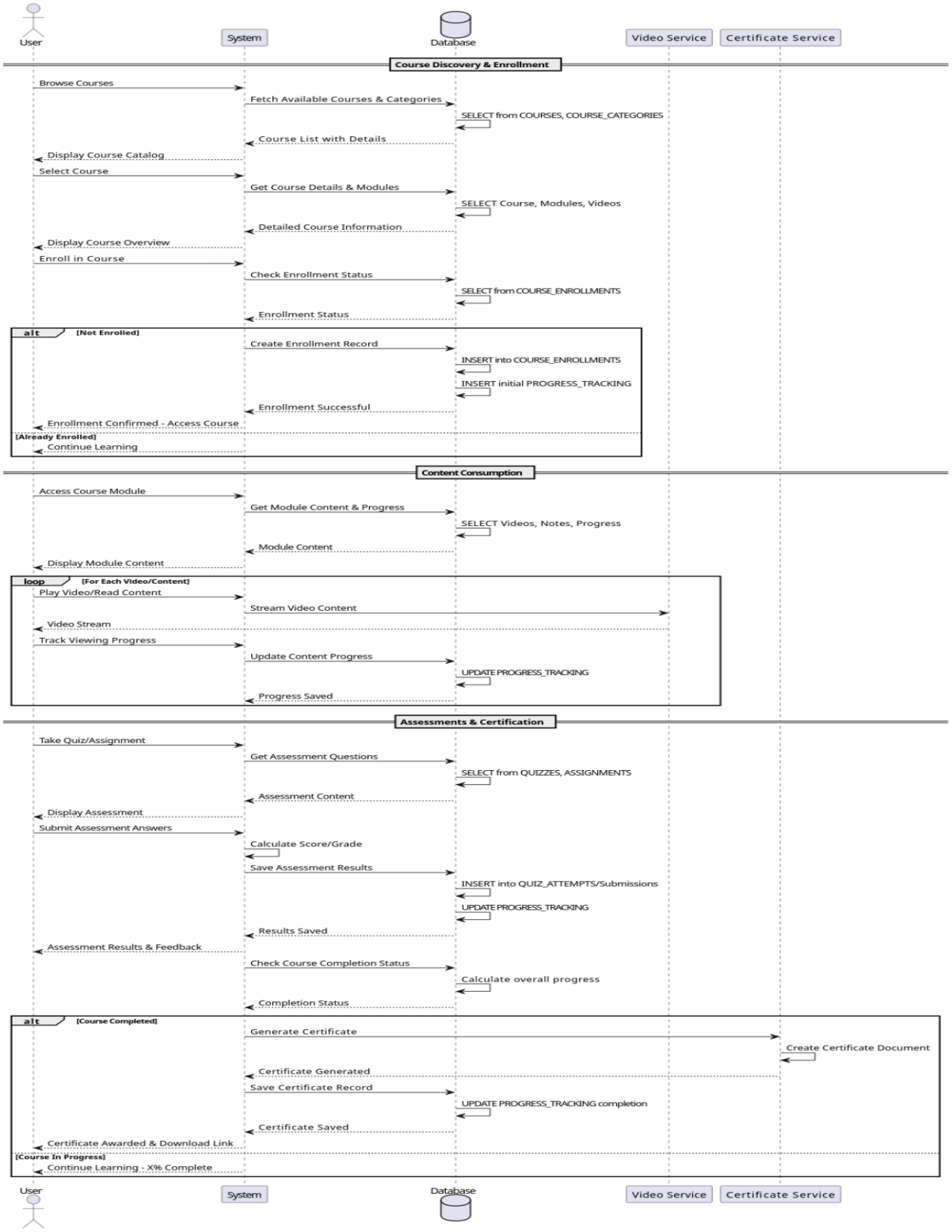

Figure 10: Course Enrollment Flow

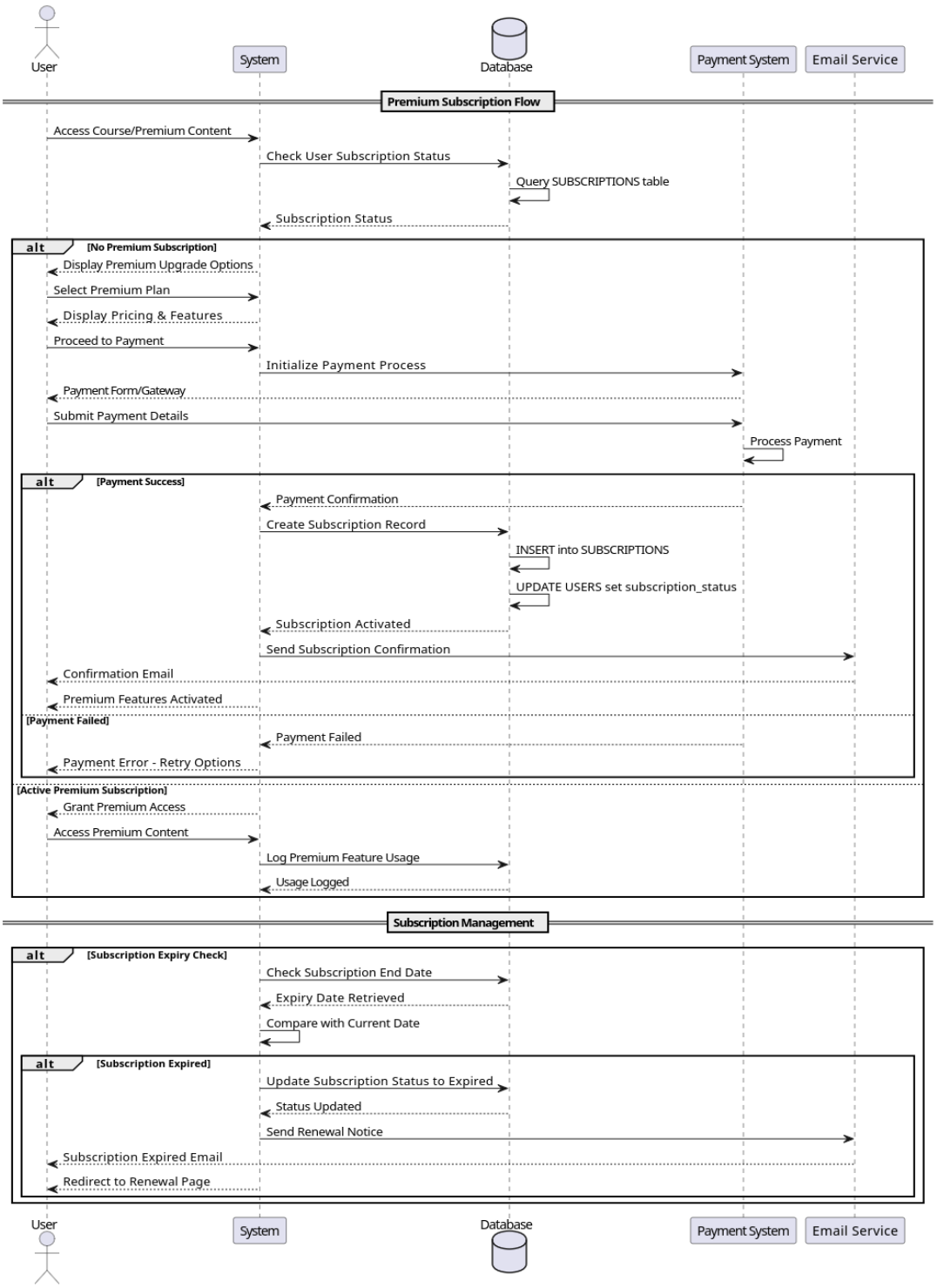

Figure 11: Premium Subscription Flow

3.2.4 Algorithm Design

Adaptive Question Difficulty Algorithm (AQDA)

Paathshala integrates an Adaptive Question Difficulty Algorithm to personalize the assessment process.The algorithm dynamically adjusts the difficulty level of MCQ questions in real time based on the learner's previous answers.If a user answers correctly, the next question becomes slightly more challenging; if answered incorrectly, an easier one is presented.

This adaptive testing method ensures balanced progression and maintains learner engagement throughout the session.It helps measure true understanding by continuously aligning question difficulty with user performance.

Algorithm Steps :

Start the test with a medium-level question.

If the user answers correctly → serve a harder question.

If the user answers incorrectly → serve an easier question.

Repeat until all questions are attempted.

Save user performance data for personalized insights.

3.2.4 User Interface Design

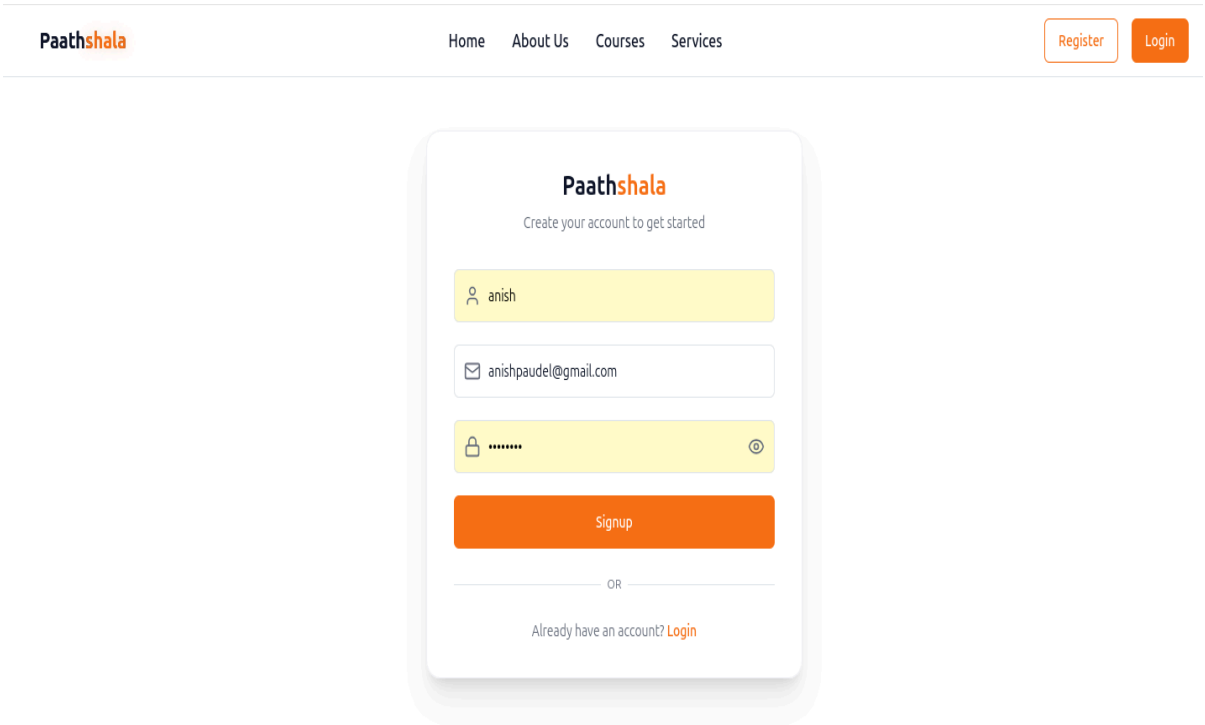

Figure 12: Signup Page

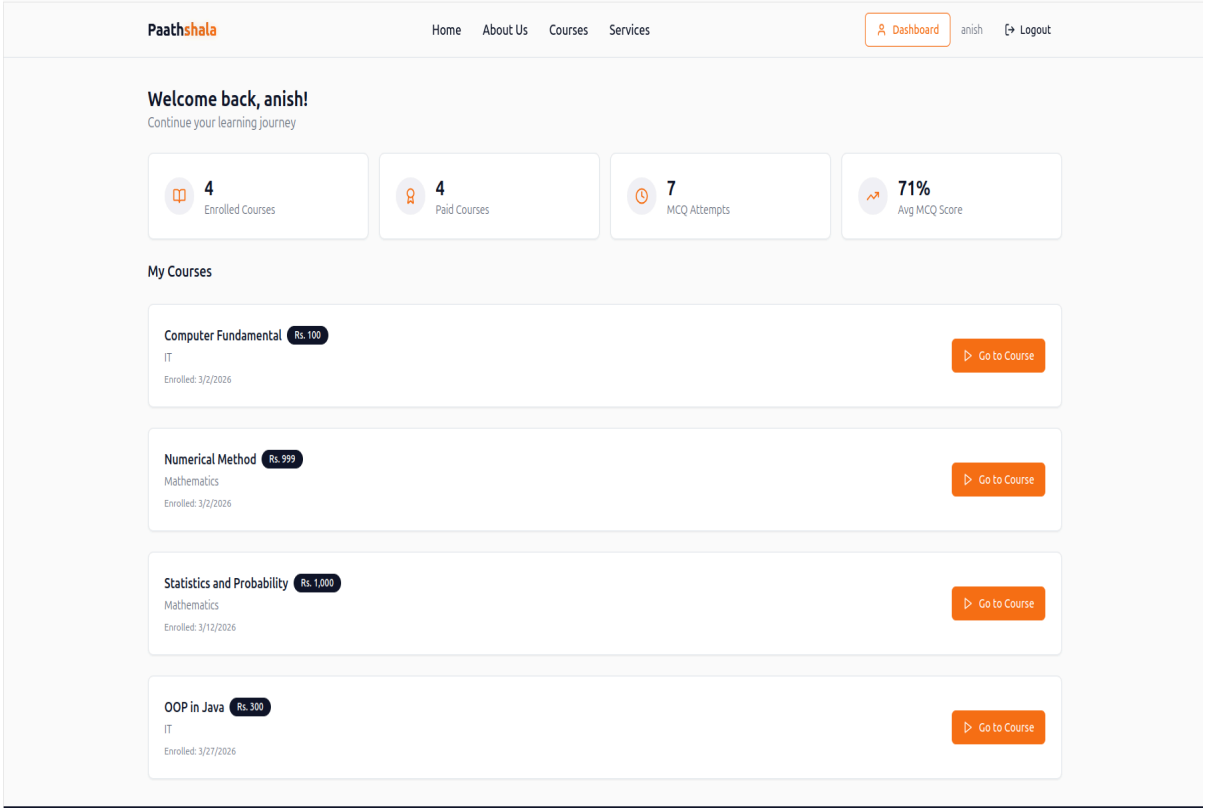

Figure 13: Student Dashboard Page

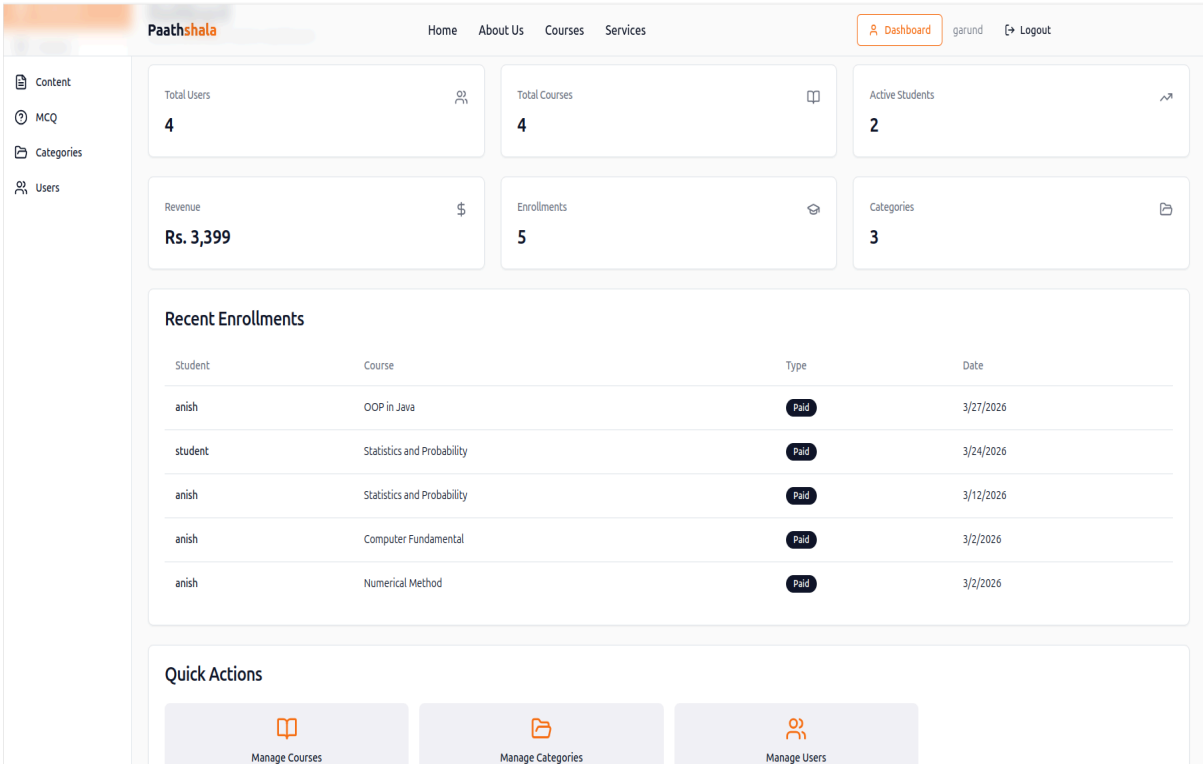

Figure 14: Admin Dashboard Page

Chapter 4: Implementation & Testing

4.1 Implementation

In this phase, the theoretical design is transformed into a functional system. With the design phase complete, the focus now shifts to technical implementation. This involves coding the application based on the system's requirements and specifications.

4.1.1 Tools Used (CASE tools, Programming languages, Database platforms)

1. Programming Languages

- **Java**: Used for developing the backend with Spring Boot and Spring Security including JWT Stateless.
- **React Js**: Used for building the interacting frontend .
- **HTML & CSS**: Used for structuring and styling the frontend.

2. Database Platforms

- **PostgreSQL**: Serves as the primary database for storing user data, media address and related information.

4. Additional Tools and Libraries

- **Docker**: Used for containerizing the application to ensure portability and consistency across environments.
- **Maven**: Used as the build automation tool for managing dependencies in the Spring Boot application.
- **Tailwind CSS**: Used for styling the frontend, enabling rapid and consistent UI development.
- **Git/Github**: Used for version control and collaborative work process.

4.2 Testing

Testing is a process of executing a program with the intent of finding an error. A good test case is one that has a high probability of finding an undiscovered error. The primary objective for test case design is to derive a set of tests that has the highest livelihood for uncovering defects in software. To accomplish this objective two different categories of test case design are used.

4.2.1 Unit Testing

Unit Testing is the first stage of the testing process where individual components of the Paathshala are tested in isolation. Each function, class, or module is examined to ensure it performs as intended. For example, user registration input validation and quiz result evaluation methods will be tested independently. Tools such as JUnit for the backend and Jest for the frontend will be used to perform these tests, ensuring the reliability of core functionalities before integration.

Example Module Tested : User Signup Module.

S.N	Procedure	Test Data	Expected Outcome	Actual Result	Test Result
1	Register user with valid field	username:anish, email:poudelanish@gmail.com, password:@123Anish	Check email for confirmation link	Check email for confirmation link	pass
2	Register user with existing username	username:anish, Email:poudelanish, Password:@1234Anish	Username already exists	Username already exists	pass

4.2.2 System Testing

System Testing evaluates the entire Paathshala as a fully integrated system. This phase verifies that the platform as a whole meets the specified requirements and functions correctly in real-world conditions. End-to-end testing will be performed, such as the complete journey of a user from registration, subscription, course enrollment, quiz attempt, and progress tracking. System testing will also involve testing the platform's performance under heavy loads when multiple users stream videos simultaneously and checking the security features, including role-based access control for users and admins. Both functional and non-functional aspects of the system will be validated during this phase, including performance, security, and usability.

S.N	Procedure	Test Data	Expected Outcome	Actual Result	Test Result
1	Enrolled to existing course	userId:1, courseId:5	Enrolled Successfully	Enrolled Successfully	pass
2	Enrolled to deleted course	userId:1, courseId:10	Course not found	Course not found	pass

4.2.3 Integration Testing

Integration Testing comes after unit testing and focuses on the interaction between different modules of the system. It ensures that components work together seamlessly and that data flows correctly across modules. In the context of the Paathshala, this will include verifying the process from user registration to email verification, the flow between the subscription module and the payment gateway with corresponding database updates, and ensuring that video content stored in file storage is accessible to users who are properly enrolled. Tools such as Postman and Spring Test will be applied to perform integration testing effectively.

4.2.4 User Acceptance Testing

User Acceptance Testing (UAT) is the final phase of the testing process. At this stage, actual users and stakeholders validate the system to ensure it fulfills business needs and expectations. Selected groups of students, both basic and premium users, as well as administrators, will interact with the platform in a controlled environment. Their feedback on usability, accessibility, and performance will be collected and analyzed. For instance, UAT will confirm whether students can easily enroll and access course materials, whether administrators can upload and manage content without errors, and whether payment scenarios work correctly. This phase ensures that the Paathshala is ready for deployment and meets the expectations of end users.

Chapter 5: Conclusion and Future Recommendation

5.1 Conclusion

The Paathshala project successfully addresses the growing need for a comprehensive, accessible, and interactive learning platform. By combining a user-friendly frontend, robust backend, and scalable infrastructure, Paathshala provides students and educators with a seamless environment for course enrollment, content delivery, and performance tracking.

The project demonstrates effective implementation of modern web technologies, including React for the frontend and a structured backend for secure data handling. Rigorous testing and deployment processes ensure reliability, stability, and smooth operation. Paathshala stands as a valuable educational tool capable of supporting both learners and administrators efficiently.

5.2 Future Recommendations

While the Paathshala project successfully meets its core objectives, there are several areas for improvement and expansion to enhance its functionality and user experience. Here are some future recommendations:

- **Mobile Application Development:** Create native or cross-platform mobile apps to provide offline access and improve accessibility for users on smartphones and tablets.
- **Gamification Features:** Implement points and rewards system to increase student motivation and engagement.
- **AI-Powered Recommendations:** Use AI to suggest courses, quizzes, and learning materials tailored to individual student performance and preferences.
- **Collaboration Tools:** Incorporate discussion forums, chat rooms, and group project features to encourage collaborative learning.
- **Content Management Improvements:** Enable bulk uploads, versioning, and content scheduling for easier management by administrators and educators.

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